

Getting Started with R

Data Analytics using R (textbook) - Chapter 2

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Working with Directory

getwd () - Get Current Working Directory

```
getwd()
```

```
[1] "C:/Users/User1/Documents/R"
```

setwd () - Set Working Directory

```
setwd("C:/path/to/my_directory")
```

On Windows, use '/' or \' as file separators.

dir() and list.files() Functions

```
dir(path = ".", pattern = NULL, all.files = FALSE, full.names = FALSE,  
    recursive = FALSE, ignore.case = FALSE, include.dirs = FALSE)
```

Example 1: List files in current directory

```
dir()
```

```
character(0)
```

Example 2: List files in a specific path

```
dir(path = "C:/Users/Seema_acharya")
```

```
[1] "AppData"           "Application Data"  
[3] "ATT_Connect_Setup.exe" "CD95F661A5C444F5A6AAECD9I1C2410a.TMP"  
[5] "Contacts"          "Cookies"  
[7] "Desktop"           "Documents"  
[9] "Downloads"         "Favorites"  
[11] "Links"             "Local Settings"  
[13] "Music"             "My Documents"  
[15] "NetHood"           "NTUSER.DAT"  
[17] "ntuser.dat.LOG1"   "ntuser.dat.LOG2"  
[19] "NTUSER.DAT{...}.TM.blf" "NTUSER.DAT{...}.TMContainer..."  
[21] "NTUSER.DAT{...}.TMContainer..." "ntuser.ini"  
[23] "ntuser.pol"        "Pictures"  
[25] "PrintHood"         "Recent"  
[27] "Saved Games"       "Searches"  
[29] "SendTo"            "Start Menu"  
[31] "Templates"         "Videos"
```

Advanced `dir()` Usage

Example 3: Full (absolute) paths

```
dir(path = "/home/anil/Documents/dsur26/programs", full.names = TRUE)
```

Example 4: Pattern matching – files starting with “D”

```
dir(path = "/home/anil/Documents/dsur26/programs", pattern = "^m")
```

```
[1] "Desktop" "Documents" "Downloads"
```

Example 5: Recursive listing

```
dir(path = "/home/anil/Documents/dsur26/programs", recursive = TRUE, include.dirs = TRUE)
```

```
[1] "db" "db/Demo.0" "db/Demo.ns" "db/local.0"
[5] "db/local.ns" "db/mongod.lock" "db/MyDB.0" "db/MyDB.ns"
```

list.files() is equivalent to *dir()* and accepts the same arguments.

Data Types in R

R Objects:

- Vector
- List
- Matrix
- Array
- Factor
- **Data Frames**

Basic Data Types

Type	Description	Example
Logical	TRUE / FALSE	TRUE, F
Numeric	Numbers	2, 76.25
Integer	Whole numbers	2L
Character	Text strings	"Data Science"
Double	Double precision	76.25
Complex	Complex numbers	5 + 5i
Raw	Raw bytes	—

Logical Data Type

```
TRUE  
class(TRUE)  
T  
class(F)  
FALSE  
class(FALSE)
```

```
[1] TRUE  
[1] "logical"  
[1] TRUE  
[1] "logical"  
[1] FALSE  
[1] "logical"
```

T is shorthand for TRUE, F for FALSE.

Numeric Data Type

```
2  
class(2)  
76.25  
class(76.25)
```

```
[1] 2  
[1] "numeric"  
[1] 76.25  
[1] "numeric"
```

Integer Data Type

Note: Use the L suffix to create an integer. Integer is a subclass of numeric.

```
2L
class(2L)
is.numeric(2)
is.numeric(2L)
is.integer(2)
is.integer(2L)
```

```
[1] 2
[1] "integer"
[1] TRUE
[1] TRUE
[1] FALSE
[1] TRUE
```

Integers are numeric, but not all numbers are integers.

Character and Double Data Types

Character

```
"Data Science"  
class("Data Science")  
is.character("Data Science")
```

```
[1] "Data Science"  
[1] "character"  
[1] TRUE
```

Double (double precision floating point)

```
typeof(76.25)
```

```
[1] "double"
```

By default, numbers are “double” unless explicitly specified as integer with L.

Complex Data Type

```
5 + 5i  
class(5 + 5i)
```

```
[1] 5+5i  
[1] "complex"
```

Variables in R

```
RectangleHeight <- 2  
RectangleWidth  <- 4  
RectangleArea   <- RectangleHeight * RectangleWidth  
RectangleArea
```

```
[1] 8
```

List objects in environment

```
ls()
```

```
[1] "RectangleArea" "RectangleHeight" "RectangleWidth"
```

Clean environment

```
rm(list = ls())  
ls()
```

```
character(0)
```

Exploring the mtcars Dataset

Background

- Extracted from the 1974 *Motor Trend* US magazine.
- Fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973–74 models).

Variables in mtcars

Variable	Description
mpg	Miles/(US) gallon
cyl	Number of cylinders
disp	Displacement (cu.in.)
hp	Gross horsepower
drat	Rear axle ratio
wt	Weight (1000 lbs)
qsec	1/4 mile time
vs	V/S
am	Transmission (0 = auto, 1 = manual)
gear	Number of forward gears
carb	Number of carburetors

Loading the mtcars Dataset

```
# Check installed packages
installed.packages()

# Load the datasets package
library(datasets)

# Display the dataset
mtcars
```

Output (first few rows):

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

summary() Command

Provides a six-point summary (min, 1st quartile, median, mean, 3rd quartile, max) for each variable.

```
summary(mtcars)
```

mpg	cyl	disp	hp
Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0
1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5
Median :19.20	Median :6.000	Median :196.3	Median :123.0
Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7
3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0
Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0
drat	wt	qsec	vs
Min. :2.760	Min. :1.513	Min. :14.50	Min. :0.0000
1st Qu.:3.080	1st Qu.:2.581	1st Qu.:16.89	1st Qu.:0.0000
Median :3.695	Median :3.325	Median :17.71	Median :0.0000
Mean :3.597	Mean :3.217	Mean :17.85	Mean :0.4375
3rd Qu.:3.920	3rd Qu.:3.610	3rd Qu.:18.90	3rd Qu.:1.0000
Max. :4.930	Max. :5.424	Max. :22.90	Max. :1.0000
am	gear	carb	
Min. :0.0000	Min. :3.000	Min. :1.000	
1st Qu.:0.0000	1st Qu.:3.000	1st Qu.:2.000	
Median :0.0000	Median :4.000	Median :2.000	
Mean :0.4062	Mean :3.688	Mean :2.812	
3rd Qu.:1.0000	3rd Qu.:4.000	3rd Qu.:4.000	
Max. :1.0000	Max. :5.000	Max. :8.000	

str() Command – Structure

Displays the internal structure of an object.

Example 1: str() on itself

```
str(str)
```

```
function(object, ...)
```

Example 2: str() on ls()

```
str(ls)
```

```
function (name, pos = -1L, envir = as.environment(pos), all.names = FALSE,  
  pattern, sorted = TRUE)
```

str() on mtcars

```
str(mtcars)
```

```
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : num  1 1 1 0 0 0 0 0 0 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```


More str() Examples

Numeric vector

```
x <- rnorm(100, 2, 4)
str(x)
```

```
num [1:100] -3.342 -5.029 0.451 2.856 4.927 ...
```

Matrix

```
m <- matrix(rnorm(100), 10, 10)
str(m)
```

```
num [1:10, 1:10] -2.231 1.089 0.573 -0.183 0.964 ...
```

```
m[, 1]
```

```
[1] -2.2310749  1.0885324  0.5730995 -0.1827884  0.9638976
[6]  1.2520684 -1.8088454  0.3247033  0.7654839 -0.3100722
```

View(), head(), tail()

View() – spreadsheet-like viewer

```
View(mtcars)
```

head() – first n observations (default 6)

```
head(mtcars, n = 6)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

head(x, n)

- x is the dataset or object.
- n specifies the number of rows to display.
- When n is a negative number, head() returns all rows except the last |n| rows.

tail() – last n observations (default 6)

```
tail(mtcars, n = 5)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4
Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6
Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2

ncol(), nrow(), edit(), fix()

ncol() – number of columns

```
ncol(mtcars)
```

```
[1] 11
```

nrow() – number of rows

```
nrow(mtcars)
```

```
[1] 32
```

edit() – edit dataset (temporary, changes not saved)

```
edit(mtcars)
```

fix() – edit dataset (changes saved permanently)

```
fix(mtcars)  
View(mtcars) # see modifications
```

Command	Result
<code>head(mtcars)</code>	Displays the first 6 rows (default).
<code>head(mtcars, 10)</code>	Displays the first 10 rows.
<code>head(mtcars, -10)</code>	Displays all rows except the last 10 rows (22 rows).
<code>tail(mtcars)</code>	Displays the last 6 rows.
<code>tail(mtcars, -10)</code>	Displays all rows except the first 10 rows (22 rows).

data() Function

Lists all available datasets in loaded packages.

```
data()
```

Load the trees dataset

```
data(trees)  
trees
```

```
   Girth Height Volume  
1    8.3     70   10.3  
2    8.6     65   10.3  
3    8.8     63   10.2  
4   10.5     72   16.4  
5   10.7     81   18.8  
6   10.8     83   19.7  
...
```

Summary of trees

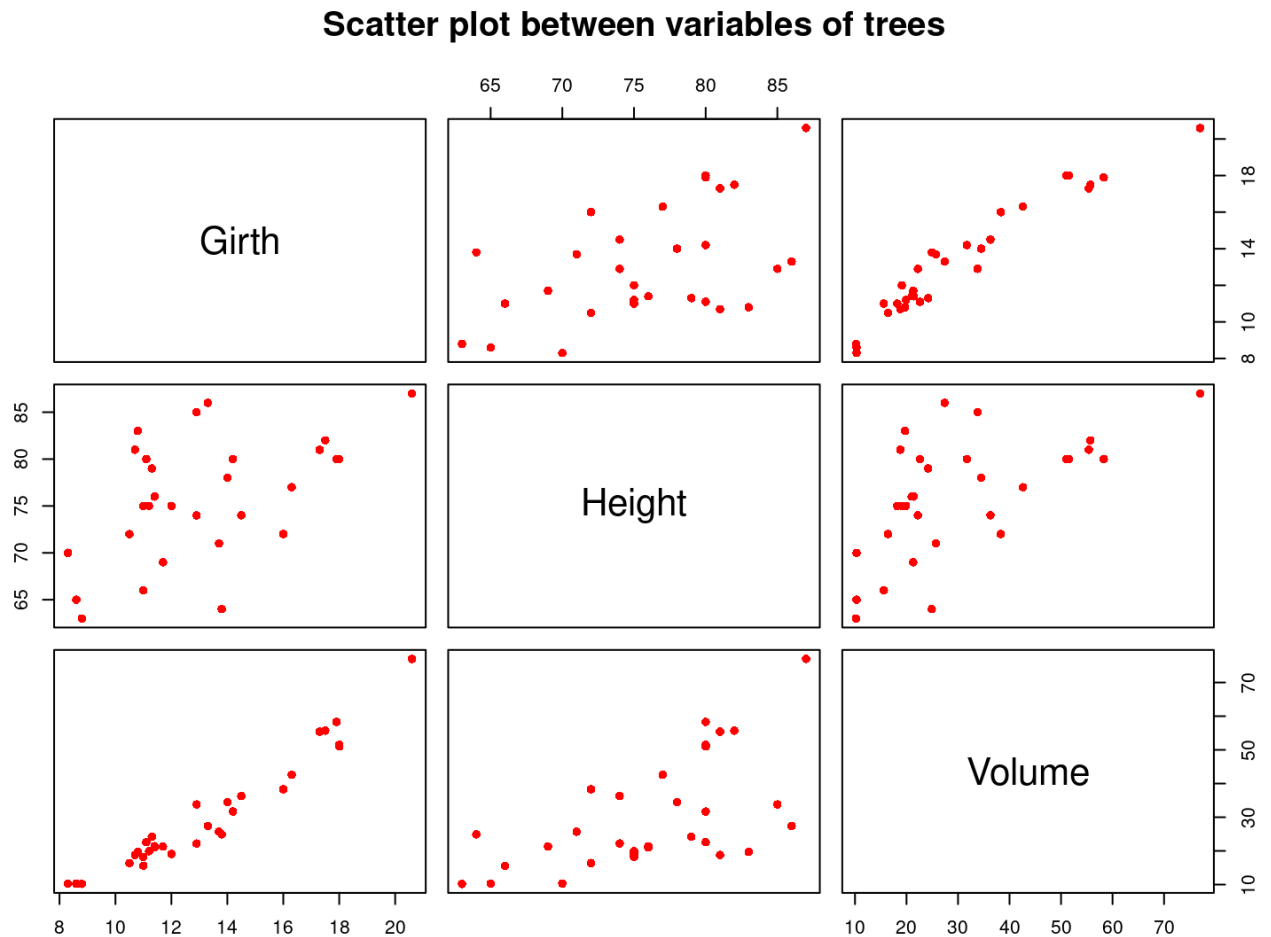
```
summary(trees)
```

Girth		Height		Volume	
Min.	: 8.30	Min.	:63	Min.	:10.20
1st Qu.	:11.05	1st Qu.	:72	1st Qu.	:19.40
Median	:12.90	Median	:76	Median	:24.20
Mean	:13.25	Mean	:76	Mean	:30.17
3rd Qu.	:15.25	3rd Qu.	:80	3rd Qu.	:37.30
Max.	:20.60	Max.	:87	Max.	:77.00

Visualizing Data with plot ()

Scatter plot of trees variables

```
plot(trees, col = "red", pch = 16,  
     main = "Scatter plot between variables of trees")
```



save.image() Function

Saves an external representation of R objects to a file.

Syntax

```
save.image(file = "RData", version = NULL, ascii = FALSE, safe = TRUE)
```

Parameters

- **file**: name of file (use .RData extension)
- **ascii**: TRUE for ASCII format, FALSE for binary (default)

Example

```
save.image(file = "myworkspace.RData")
```

To load later:

```
load("myworkspace.RData")
```

Getting Help in R

Use `?` or `help()`.

```
?summary  
help(summary)  
?rnorm  
help(rnorm)
```


Key Differences

`head()` vs `tail()`

Feature	<code>head()</code>	<code>tail()</code>
Purpose	Shows records from start	Shows records from end
Default n	6	6

`edit()` vs `fix()`

Feature	<code>edit()</code>	<code>fix()</code>
Changes saved?	No (temporary)	Yes (permanent)
Assignment needed?	Yes	No

`class()` vs `typeof()`

Feature	<code>class()</code>	<code>typeof()</code>
Purpose	Object class/type	Internal data type
Granularity	Higher level	Lower level

Exercise: BOD Dataset

BOD (Biochemical Oxygen Demand) dataset:

	Time	demand
1	1	8.3
2	2	10.3
3	3	19.0
4	4	16.0
5	5	15.6
6	7	19.8

Question 1

What will `nrow(BOD)` return?

```
nrow(BOD)
```

```
[1] 6
```

Question 2

What will `head(BOD, n = 3)` display?

```
head(BOD, n = 3)
```

	Time	demand
1	1	8.3
2	2	10.3
3	3	19.0

Exercise: Predict the Output

(a) `summary(mtcars$mpg)`

```
summary(mtcars$mpg)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
10.40	15.43	19.20	20.09	22.80	33.90

(b) `summary(c(3,2,1,2,4,6))`

```
summary(c(3,2,1,2,4,6))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.000	2.000	2.500	3.000	3.750	6.000

(c) `str(c(1,2,3,4))`

```
str(c(1,2,3,4))
```

```
num [1:4] 1 2 3 4
```

Exercise: Predict the Output (continued)

(d) `str(c("Mon", "Tue", "Wed", "Thurs"))`

```
str(c("Mon", "Tue", "Wed", "Thurs"))
```

```
chr [1:4] "Mon" "Tue" "Wed" "Thurs"
```

(e) `head(c("Mon", "Tue", "Wed", "Thurs"), 2)`

```
head(c("Mon", "Tue", "Wed", "Thurs"), 2)
```

```
[1] "Mon" "Tue"
```

(f) `tail(c("Mon", "Tue", "Wed", "Thurs"), 2)`

```
tail(c("Mon", "Tue", "Wed", "Thurs"), 2)
```

```
[1] "Wed" "Thurs"
```

(g) `class(76.25L)`

```
class(76.25L)
```

```
[1] "integer"
```

(The L suffix forces the number to be integer.)

Summary

- **Directory commands:** `getwd()`, `setwd()`,
`dir()/list.files()`
- **Data types:** Logical, Numeric, Integer, Character, Double, Complex, Raw
- Use `class()` to check type; `typeof()` for more specific type.
- **Data exploration functions:**
 - *`summary()` – six-point summary*
 - *`str()` – internal structure*
 - *`View()` – spreadsheet viewer*
 - *`head()` / `tail()` – first / last rows*
 - *`nrow()` / `ncol()` – dimensions*
 - *`edit()` / `fix()` – edit data*
- **Environment:** `ls()`, `rm()`, `save.image()`

Key Terms

Term	Description
<code>dir()</code>	Returns character vector of file/directory names
<code>getwd()</code>	Returns absolute file path of current working directory
<code>setwd()</code>	Changes current working directory
<code>typeof()</code>	Checks the data type of an object
<code>class()</code>	Reveals the data type/class of an object
<code>summary()</code>	Provides statistical summary of a dataset
<code>str()</code>	Displays internal structure of an object
<code>ls()</code>	Lists all objects in working environment
<code>data()</code>	Lists all available datasets

